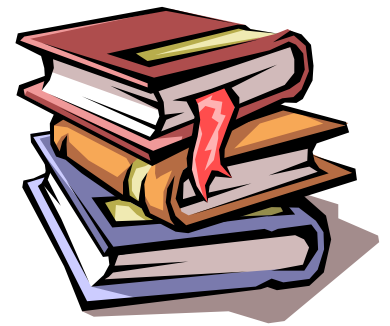




TTK4150 Nonlinear Control Systems

Syllabus/Reading list

The syllabus of the course is the material presented at the lectures and in the assignments. This is covered by the lecture handouts, the assignments and their solution manuals, and the literature below:

Textbook: H.K. Khalil: *Nonlinear Systems*

Topics	Chapters	Comments
Important mathematical results used throughout the course	Appendix A	
Nonlinear phenomena	Chapter 1	
Fundamental properties	Chapter 3: Section 3.1	
Phase plane analysis	Chapter 2: Sections 2.1 – 2.6	Bifurcations and chaos theory are not included in the syllabus
Lyapunov stability	Chapter 4: Sections 4.1 – 4.5 Section 4.7: Theorem 4.15 Corollary 4.3 Chapter 8: Sections 8.2 – 8.3	• Theorem 4.3, is additional material • Example 4.11 is additional material • Pages 133 to 135 - end of example 4.12, are background material • Example 8.10 after “$c=1$” is additional material • Pages 325-329 starting with Theorem 8.5 are additional material
Input-to-State Stability	Chapter 4: Section 4.9	Background material is given in paper and talk by Sontag and mini-course by Loria.
Input-Output Stability	Chapter 5: Sections 5.1 and 5.4	Sections 5.2 and 5.3 are additional material. Example 5.14 is additional material
Passivity	Chapter 6: Sections 6.1 – 6.2 Sections 6.4 – 6.5	<i>Input-feedforward passivity</i> and <i>output-feedback passivity</i> are additional material. Section 6.3 is additional material. Pages 254 – 259 including Example 6.12 are additional material.
Passivity-based control design	Chapter 14: Section 14.4	Some material on passivity of linear systems (from Lozano et al.) is given in the lecture handouts and assignments and is thus part of the syllabus.
Input-Output linearization	Chapter 13: Sections 13.1, 13.2 and 13.4	Example 13.16 – page 538 is additional material.
Adaptive control	Compendium from Slotine & Li’s book: Chapter 8: Pages 311-316 Section 8.2 pp. 326-330, p.	

	333 Example 8.4 pp. 334-335 Chapter 9: Section 9.2.1 Lavretsky and Wise: Robust and Adaptive Control Chapter 7	
Backstepping	Chapter 14: Section 14.3	Pages 589 – 598 (until end of example 14.12)

Background material:

E. D. Sontag: “The ISS Philosophy as a Unifying Framework for Stability-Like Behavior”, in *Nonlinear Control in the Year 2000, Vol 2*. Eds. A. Isidori, F. Lamnabhi-Lagarrigue and W. Respondek, Springer Lecture Notes in Control and Information Sciences 258, 2001, pp. 443-468

E. D. Sontag: “The ISS Philosophy as a Unifying Framework for Stability-Like Behavior”, Bode Talk (with narrative) at 2002 IEEE Conference on Decision and Control.

A. Loria: “Cascaded nonlinear time-varying systems: analysis and design”, Minicourse at the “Congreso Internaional de Computation”, Mexico, Nov. 14-16, 2001.

Lozano, Brogliato, & Egeland and Maschke: Dissipative Systems Analysis and Control, Springer Verlag, Section 2.3

Additional material: This is course material that you will not get direct questions about at the exams, but which you should have read such that you know that it exists and where to find it in case you need it later.

Background material: This will neither be topics for direct questions at the exams, but is information that may provide valuable insight to really understand the core material of the course.